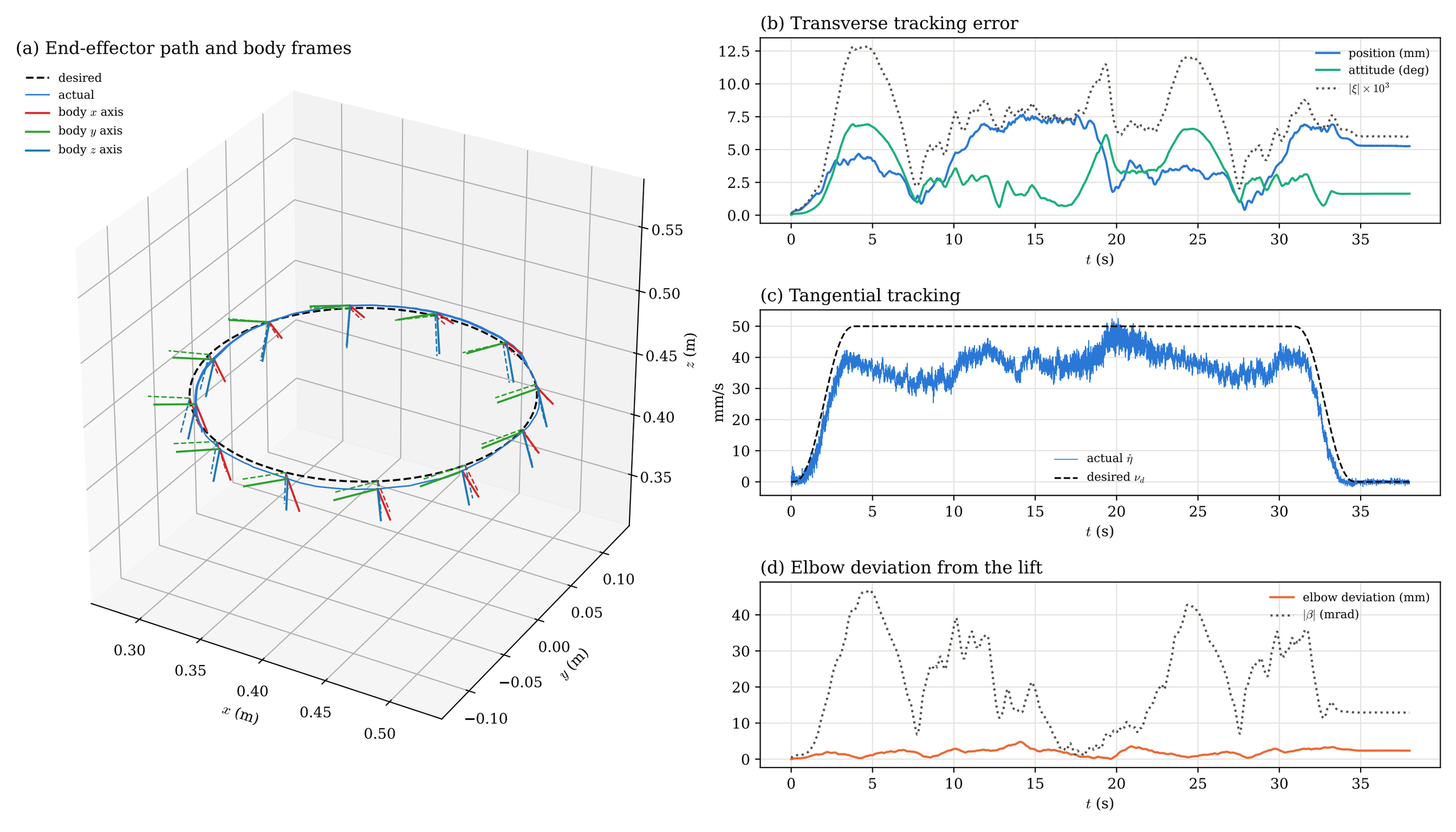

(a) End-effector path and body frames
desired
actual
body x axis
body y axis
body z axis
0.55
0.50
0.45
0.40
0.35
z (m)
0.10
0.05
0.00
−0.05
−0.10
y (m)
0.30
0.35
0.40
0.45
0.50
x (m)
(b) Transverse tracking error
position (mm)
attitude (deg)
$|\xi| \times 10^3$
12.5
10.0
7.5
5.0
2.5
0.0
0
5
10
15
20
25
30
35
t (s)
(c) Tangential tracking
mm/s
50
40
30
20
10
0
actual $\dot{\eta}$
desired ν_d
0
5
10
15
20
25
30
35
t (s)
(d) Elbow deviation from the lift
elbow deviation (mm)
$|\beta|$ (mrad)
40
30
20
10
0
0
5
10
15
20
25
30
35
t (s)